Discrete diffusion models: From continuous to categorical

Pipi Hu

Beijing Institute of Mathematical Sciences and Applications

October 8, 2025

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Feynman's words

What I cannot create, I do not understand.

Outline

1. From Continuous to Discrete: The Challenge
2. Discrete Diffusion: Mathematical Foundation
3. Score-based Discrete Diffusion
4. Autoregressive vs Diffusion for Discrete Data
5. Implementation Strategies
6. Advanced Discrete Diffusion
7. Applications and Examples
8. Theoretical Analysis
9. Challenges and Open Problems
10. Conclusion

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

The discrete data challenge

Traditional diffusion models work well for continuous data, but many real-world applications involve discrete data:

- Text generation: Words, tokens, characters
- Image generation: Pixel values, categorical features
- Molecular generation: Atom types, bond types
- Music generation: Notes, chords, instruments

Key question: How to apply diffusion principles to discrete/categorical data?

Why not just discretize continuous diffusion?

Direct discretization of continuous diffusion faces several challenges:

1. Score function undefined: $\nabla \log p(x)$ doesn't exist for discrete x
2. Noise structure: Gaussian noise doesn't preserve categorical structure
3. Sampling complexity: Discrete spaces have different geometry
4. Training instability: Continuous losses don't translate directly

Need: New mathematical framework for discrete diffusion

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Outline

1. From Continuous to Discrete: The Challenge
2. Discrete Diffusion: Mathematical Foundation
3. Score-based Discrete Diffusion
4. Autoregressive vs Diffusion for Discrete Data
5. Implementation Strategies
6. Advanced Discrete Diffusion
7. Applications and Examples
8. Theoretical Analysis
9. Challenges and Open Problems
10. Conclusion

Discrete state spaces and transition matrices

For discrete data $x \in \{1, 2, \ldots, K\}$, we need discrete transition probabilities.
Forward process: Define transition matrix $\mathbf{Q}_t$ such that

$$q(x_t | x_{t-1}) = \mathbf{Q}_t[x_{t-1}, x_t] \tag{1}$$

Key properties:

- $\mathbf{Q}_t$ is a stochastic matrix: $\sum_{x_t} \mathbf{Q}_t[x_{t-1}, x_t] = 1$
- $\mathbf{Q}_t[x_{t-1}, x_t] \geq 0$ for all x_{t-1}, x_t
- Can be time-dependent: $\mathbf{Q}_t$ changes with t

Discrete forward process

The discrete forward process is defined as:

$$q(x_t|x_0) = \sum_{x_1,\ldots,x_{t-1}} \prod_{s=1}^{t} \mathbf{Q}_s[x_{s-1}, x_s] \tag{2}$$

For uniform transitions (common choice):

$$\mathbf{Q}_t = (1-\beta_t)\mathbf{I} + \beta_t \mathbf{1}\mathbf{1}^T/K \tag{3}$$

where $\mathbf{I}$ is identity matrix and $\mathbf{1}\mathbf{1}^T$ is uniform transition matrix.
Interpretation: With probability $(1-\beta_t)$, stay the same; with probability β_t, jump to any state uniformly.

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Discrete reverse process

The reverse process requires learning transition probabilities:

$$p_\theta(x_{t-1}|x_t) = \text{Cat}(x_{t-1}; \boldsymbol{\pi}_\theta(x_t, t)) \tag{4}$$

where $\boldsymbol{\pi}_\theta(x_t, t) \in \mathbb{R}^K$ is a probability vector.
Training objective: Learn to reverse the forward process

$$\mathcal{L} = \mathbb{E}_{t,x_0,x_t}\left[-\log p_\theta(x_{t-1}|x_t)\right] \tag{5}$$

Challenge: How to parameterize $\boldsymbol{\pi}_\theta(x_t, t)$ effectively?

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Outline

1. From Continuous to Discrete: The Challenge
2. Discrete Diffusion: Mathematical Foundation
3. Score-based Discrete Diffusion
4. Autoregressive vs Diffusion for Discrete Data
5. Implementation Strategies
6. Advanced Discrete Diffusion
7. Applications and Examples
8. Theoretical Analysis
9. Challenges and Open Problems
10. Conclusion

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Discrete score function

For discrete data, we define the discrete score function:

$$s(x,t) = \nabla \log p(x,t) = \frac{\partial \log p(x,t)}{\partial x} \tag{6}$$

Problem: This is not well-defined for categorical x.
Solution: Use logits instead of direct gradients:

$$s(x,t) = \log p(x,t) - \log \sum_{x'} p(x',t) \tag{7}$$

Or equivalently:

$$s(x,t) = \log p(x,t) + C \tag{8}$$

where C is a constant that doesn't affect the relative probabilities.

Discrete denoising score matching

For discrete data, the DSM loss becomes:

$$\mathcal{L}_{DSM} = \mathbb{E}_{t,x_0,x_t}\left[\|s_\theta(x_t,t) - \nabla_{x_t}\log q(x_t|x_0)\|^2\right] \tag{9}$$

Key insight: The score $\nabla_{x_t}\log q(x_t|x_0)$ can be computed analytically for discrete transitions. For uniform transitions:

$$\nabla_{x_t}\log q(x_t|x_0) = \frac{\mathbf{Q}_t[x_0,x_t]}{\sum_{x'}\mathbf{Q}_t[x_0,x']} - \frac{1}{K} \tag{10}$$

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Outline

1. From Continuous to Discrete: The Challenge
2. Discrete Diffusion: Mathematical Foundation
3. Score-based Discrete Diffusion
4. Autoregressive vs Diffusion for Discrete Data
5. Implementation Strategies
6. Advanced Discrete Diffusion
7. Applications and Examples
8. Theoretical Analysis
9. Challenges and Open Problems
10. Conclusion

Autoregressive models for discrete data

Traditional approach for discrete data:

$$p(x) = \prod_{i=1}^{n} p(x_i | x_{<i}) \tag{11}$$

Advantages:

- Natural for discrete data
- Easy to train with cross-entropy loss
- Can model complex dependencies

Disadvantages:

- Sequential generation (slow)
- Order dependence
- Hard to parallelize

Discrete diffusion advantages

Discrete diffusion offers several advantages over autoregressive models:

1. Parallel generation: All positions can be generated simultaneously
2. Order independence: No fixed generation order
3. Flexible conditioning: Easy to condition on partial information
4. Unified framework: Same principles as continuous diffusion

Trade-off: More complex training but more flexible generation

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Outline

1. From Continuous to Discrete: The Challenge
2. Discrete Diffusion: Mathematical Foundation
3. Score-based Discrete Diffusion
4. Autoregressive vs Diffusion for Discrete Data
5. Implementation Strategies
6. Advanced Discrete Diffusion
7. Applications and Examples
8. Theoretical Analysis
9. Challenges and Open Problems
10. Conclusion

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Discrete diffusion implementation

Strategy 1: Direct parameterization

$$p_\theta(x_{t-1}|x_t) = \mathrm{softmax}(f_\theta(x_t, t)) \tag{12}$$

where $f_\theta : \mathbb{R}^K \times \mathbb{R} \to \mathbb{R}^K$

Strategy 2: Score-based approach

$$s_\theta(x_t, t) = f_\theta(x_t, t) - \log \sum_{x'} \exp(f_\theta(x', t)) \tag{13}$$

Strategy 3: Embedding approach

- Embed discrete states in continuous space
- Apply continuous diffusion in embedding space
- Project back to discrete space

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Training discrete diffusion models

Training procedure:

1. Sample $x_0 \sim p_{data}$
2. Sample $t \sim \text{Uniform}(0, T)$
3. Sample $x_t \sim q(x_t|x_0)$ using forward process
4. Train model to predict x_0 or x_{t-1} from x_t

Loss functions:

- Cross-entropy: $\mathcal{L} = -\log p_\theta(x_0|x_t)$
- Score matching: $\mathcal{L} = \|s_\theta(x_t, t) - s_{true}(x_t, t)\|^2$
- Variational bound: $\mathcal{L} = \text{ELBO}(x_0, x_t)$

北京雁栖湖应用数学研究院 Beijing Institute of Mathematical Sciences and Applications

Outline

1. From Continuous to Discrete: The Challenge
2. Discrete Diffusion: Mathematical Foundation
3. Score-based Discrete Diffusion
4. Autoregressive vs Diffusion for Discrete Data
5. Implementation Strategies
6. Advanced Discrete Diffusion
7. Applications and Examples
8. Theoretical Analysis
9. Challenges and Open Problems
10. Conclusion

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Categorical diffusion

For categorical data with K classes, use categorical distribution:

$$q(x_t|x_{t-1}) = \mathrm{Cat}(x_t; \boldsymbol{\alpha}_t) \tag{14}$$

where $\boldsymbol{\alpha}_t$ is the concentration parameter.
Forward process:

$$\boldsymbol{\alpha}_t = (1 - \beta_t)\mathbf{e}_{x_{t-1}} + \beta_t \mathbf{1}/K \tag{15}$$

where $\mathbf{e}_{x_{t-1}}$ is the one-hot vector for x_{t-1}.

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Discrete diffusion with structure

For structured discrete data (e.g., molecules, graphs):

- Graph diffusion: Diffuse over graph structure
- Hierarchical diffusion: Multi-level discrete states
- Conditional diffusion: Condition on partial structure

Example: Molecular generation

$$p(\text{molecule}) = \prod_{t=1}^{T} p(\text{atom}_t | \text{atoms}_{<t}, \text{bonds}_{<t}) \tag{16}$$

Challenge: Maintaining chemical validity during diffusion

北京雁栖湖应用数学研究院 Beijing Institute of Mathematical Sciences and Applications

Outline

1. From Continuous to Discrete: The Challenge
2. Discrete Diffusion: Mathematical Foundation
3. Score-based Discrete Diffusion
4. Autoregressive vs Diffusion for Discrete Data
5. Implementation Strategies
6. Advanced Discrete Diffusion
7. Applications and Examples
8. Theoretical Analysis
9. Challenges and Open Problems
10. Conclusion

北京雁栖湖
应用数学研究院
Beijing Institute
of Mathematical
Sciences and Applications

Text generation with discrete diffusion

Token-level diffusion:

- Forward: Gradually replace tokens with noise
- Reverse: Learn to denoise and recover original text
- Training: Predict original tokens from noisy versions

Advantages over autoregressive:

- Parallel generation
- Better long-range dependencies
- More flexible conditioning

Image generation with discrete diffusion

Pixel-level discrete diffusion:

- Treat pixel values as discrete categories
- Forward: Gradually randomize pixel values
- Reverse: Learn to recover original pixels

Hybrid approaches:

- Continuous diffusion for low-level features
- Discrete diffusion for high-level structure
- Multi-scale discrete diffusion

Outline

1. From Continuous to Discrete: The Challenge
2. Discrete Diffusion: Mathematical Foundation
3. Score-based Discrete Diffusion
4. Autoregressive vs Diffusion for Discrete Data
5. Implementation Strategies
6. Advanced Discrete Diffusion
7. Applications and Examples
8. Theoretical Analysis
9. Challenges and Open Problems
10. Conclusion

北京雁栖湖
应用数学研究院
Beijing Institute
of Mathematical
Sciences and Applications

Discrete diffusion convergence

Theorem: Under certain conditions, discrete diffusion converges to the data distribution.

Key assumptions:

1. Forward process is ergodic
2. Reverse process is properly parameterized
3. Training loss is minimized

Proof sketch:

$$p_\theta(x_0) = \int p_\theta(x_0|x_T)p(x_T)dx_T \tag{17}$$

As $T \to \infty$, $p(x_T)$ approaches uniform distribution, and $p_\theta(x_0|x_T)$ learns to recover the data distribution.

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Discrete vs continuous diffusion

Comparison:

- Continuous: Smooth interpolation, gradient-based
- Discrete: Categorical jumps, probability-based

Mathematical differences:

- Score function: $\nabla \log p(x)$ vs $\log p(x)$
- Noise: Gaussian vs categorical
- Sampling: Langevin dynamics vs discrete transitions

Unified view: Both can be seen as special cases of a general diffusion framework

Outline

1 From Continuous to Discrete: The Challenge
2 Discrete Diffusion: Mathematical Foundation
3 Score-based Discrete Diffusion
4 Autoregressive vs Diffusion for Discrete Data
5 Implementation Strategies
6 Advanced Discrete Diffusion
7 Applications and Examples
8 Theoretical Analysis
9 Challenges and Open Problems
10 Conclusion

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Current challenges in discrete diffusion

1. Training stability: Discrete losses can be unstable
2. Sampling efficiency: Discrete sampling can be slow
3. Model capacity: Need large models for complex discrete data
4. Evaluation metrics: How to measure quality of discrete generation?

Open problems:

- Optimal transition matrices for different data types
- Theoretical guarantees for discrete diffusion
- Efficient sampling algorithms
- Better loss functions for discrete data

Future directions

Research directions:

1. Hybrid models: Combine continuous and discrete diffusion
2. Structured diffusion: Incorporate domain knowledge
3. Efficient algorithms: Faster training and sampling
4. Theoretical understanding: Better mathematical foundations

Applications:

- Scientific discovery (molecules, materials)
- Creative applications (music, art)
- Natural language processing
- Code generation

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Outline

1. From Continuous to Discrete: The Challenge
2. Discrete Diffusion: Mathematical Foundation
3. Score-based Discrete Diffusion
4. Autoregressive vs Diffusion for Discrete Data
5. Implementation Strategies
6. Advanced Discrete Diffusion
7. Applications and Examples
8. Theoretical Analysis
9. Challenges and Open Problems
10. Conclusion

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications

Summary

Discrete diffusion models provide a powerful framework for generating discrete data:

- Mathematical foundation: Discrete transition matrices and score functions
- Implementation strategies: Direct parameterization, score-based, embedding-based
- Applications: Text, images, molecules, and more
- Advantages: Parallel generation, flexible conditioning, unified framework

Key insight: Discrete diffusion extends the power of continuous diffusion to categorical data while maintaining the core principles of iterative denoising.

Questions

If you are interested in the questions, please feel free to write an email to me (hpp@bimsa.cn) with your comments.

1. How can we design optimal transition matrices for specific types of discrete data (e.g., text, molecules, graphs)?
2. What are the theoretical guarantees for convergence of discrete diffusion models?
3. How can we efficiently sample from discrete diffusion models while maintaining quality?
4. What are the best practices for training discrete diffusion models on large-scale datasets?

Welcome inputs

Any comments?

Welcome your inputs!